Assignment: Digital Clock Using JavaScript

By: Vishal Singh

Project Objective & Description

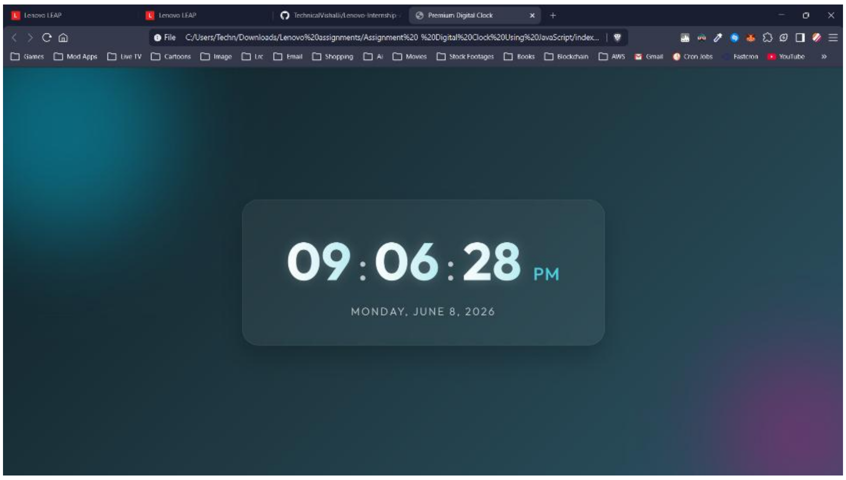
Objective: To understand how to use JavaScript to display and update real-time data on a webpage by working with the Date object, DOM manipulation, and timing functions like setInterval().

Description: In this project, I created a Digital Clock that displays the current time on a webpage, updating automatically every second without requiring a page refresh. The clock features a premium, modern design utilizing glassmorphism and dynamic background animations. It showcases hours, minutes, and seconds in a 12-hour format, along with an AM/PM indicator and the current date.

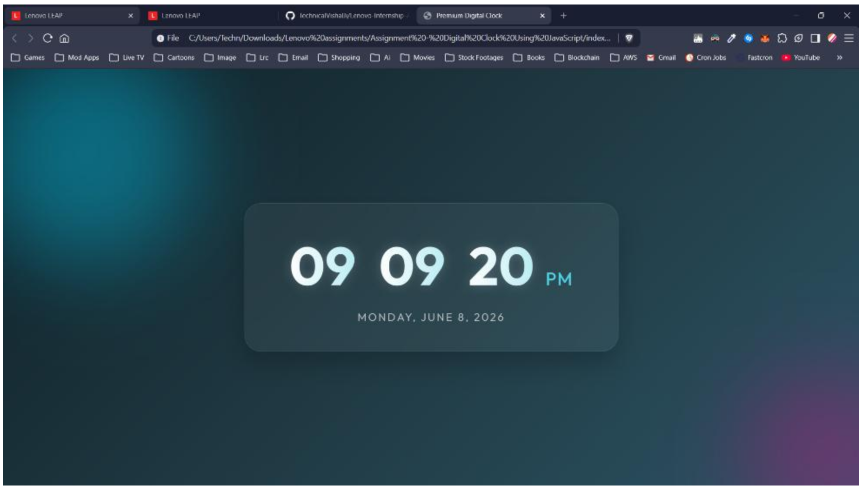
Technologies Used

- **HTML5**: Used to create the semantic structure of the digital clock, organizing elements such as hours, minutes, seconds, and date.
- **CSS3**: Used to design a visually striking, premium user interface. Techniques applied include glassmorphism (backdrop filters), smooth linear gradients, custom web fonts (Google Fonts 'Outfit'), and CSS animations for the blinking colon and moving background elements.
- **JavaScript (ES6)**: Used to fetch the real-time system time via the Date object, calculate hours, minutes, and seconds, and seamlessly update the Document Object Model (DOM) continuously using the setInterval() function.

Screenshots with Explanation



Explanation: This screenshot showcases the initial view of the Digital Clock. The clock seamlessly displays the correct system time with a blinking colon to signify real-time ticking. The premium glassmorphic container adds depth over the animated gradient background.



Explanation: This screenshot demonstrates the dynamic nature of the webpage, capturing a different time, and showing the live-updating AM/PM tracker along with the accurately formatted current date string located at the bottom of the clock container.

Project Links

GitHub Repository: Link

Conclusion & Summary

Summary of Outcomes: The Digital Clock was successfully built and operates precisely as expected, ticking every second without the need to reload the page. The interface was significantly enhanced to provide a modern, engaging aesthetic.

Key Learnings:

- Mastered the JavaScript Date object to extract specific time and date components.
- Gained practical experience with the setInterval() function to execute asynchronous code periodically, simulating a real-time clock.
- Improved DOM manipulation skills by dynamically changing the text content of specific HTML elements (document.getElementById()).
- Learned how to integrate dynamic UI components effectively without sacrificing application performance.